

Auto-Dispatch Agents — Product Context

Reference document for Claude · Prototype state as of April 2026

Overview

Auto-Dispatch is an MSP-platform feature that uses AI agents to automate three layers of ticket dispatching: routing tickets to the right team, assigning them to the right technician, and (optionally) scheduling them on that tech's calendar. Admins create one or more agents, each configured for a specific mode and scope. Each agent monitors a board and acts on tickets that meet its criteria.

There are **three agent modes**, layered as tiers — each mode includes everything the previous tier does, and adds more autonomy:

Tier	Mode	Adds
1	Route	Reads incoming tickets and moves them to the right team's board.
2	Assign	Picks the right technician on a team based on workload/availability.
3	Assign + Schedule	Picks a tech <i>and</i> places the work on their calendar.

Shared concepts (all modes)

Agent scope

Every agent is scoped to:

- A **board** it monitors (the "Dispatch from" board / intake board for Route).
- A set of ticket **statuses** it operates on, with **Include only** or **Exclude** semantics.
- An **active toggle** — agents can be paused without deletion.
- A **team** (Assign / Assign + Schedule only — Route is platform-wide).

Ticket context the agent receives

Regardless of mode, every agent has access to the following ticket fields when making a decision. Prompts should be written in plain natural language — the agent sees the resolved values, not the variable names.

- Contact company name (`{{company_name}}`)
- Contact company type (`{{company_type}}`)
- Ticket contact name (`{{contact_name}}`)
- Ticket contact types — ConnectWise PSA (`{{contact_type}}`)
- Ticket summary (`{{summary}}`)
- Ticket chat history (`{{ticket_chat}}`)
- Ticket priority (`{{priority}}`)
- Ticket type (`{{type}}`)
- Ticket subtype (`{{subtype}}`)
- Ticket item (`{{item}}`)
- Ticket agreement name (`{{ticket_agreement}}`)
- Ticket agreement type (`{{ticket_agreement_type}}`)
- Ticket configuration array (`{{ticket_configuration}}`)

Agent autonomy (all modes)

- **Limit agent workload:** when on, the agent only handles a configurable **percentage** of incoming tickets (10–100%); the rest are left for manual review.

Mode 1 — Route

Purpose: Triage agent. Watches an intake board and forwards each ticket to the most appropriate team's board.

CONFIGURATION

- **Dispatch from:** the single intake board the agent monitors.
- **Status filter:** which statuses count.
- **Routing rules:** a list of rules, one per destination board. Each rule has:
 - **Destination board** ("Dispatch to {board}").
 - **Prompt** — free-form natural-language instructions describing when this board should be chosen. The agent uses this *in addition to* the ticket context above.
 - *(Hidden for now, code preserved)* A structured `When...` condition editor that filters by Company, Company type, Type, Subtype, Item, Agreement with `is / isNot / isBlank / isNotBlank` operators. Currently commented out — only the natural-language prompt is exposed.
- **Add dispatch route** button to add more rules.

DECISIONING MODEL

The agent reads all rules together and picks the best-fit destination board for each ticket, weighing each rule's prompt against the ticket's context.

CONSTRAINTS

- Only **one Route agent** can exist at a time across the workspace.
- A routing rule **can't route to the same board it's routing from**.
- Each destination board can appear in at most one rule on the same agent.

Mode 2 — Assign

Purpose: Once a ticket lands on a team's board (manually or via a Route agent), an Assign agent picks the right technician on that team.

CONFIGURATION (EVERYTHING IN ROUTE'S "SCOPE" PLUS)

- **Team** — which team this agent serves.
- **Assignment logic:**
 - **Priority:** Highest first / Lowest first.
 - **Ticket age:** Oldest first / Newest first.
 - **Tech selection:** Most open time / Fewest tickets assigned / Round robin.
 - **Max active tickets per tech:** numeric cap; when a tech hits it, they're skipped.
 - *(Coming soon)* SLA risk, Skills match, Client familiarity.

DECISIONING MODEL

For each eligible ticket, the agent ranks technicians on the team using the assignment logic, respecting per-tech capacity, and assigns the ticket.

CONSTRAINTS

- A team can have **at most one** Assign or Assign + Schedule agent (not both).
- Currently assumes **1 team = 1 board**. Shared boards across teams are not yet supported.

Mode 3 — Assign + Schedule

Purpose: Everything Assign does, plus places the work on the assigned technician's calendar at a real time slot.

CONFIGURATION (EVERYTHING IN ASSIGN PLUS A "WORKING HOURS AND SCHEDULING" SECTION)

- **Working hours:** start/end times for the team's day (workspace timezone).
- **Schedule offset:** how far ahead from "now" to start scheduling (e.g. 60-min offset means 8:15 AM "now" → earliest slot is 9:15 AM). Options: 15/30/60 min.
- **Schedule interval:** time-slot granularity (15/30/60 min).
- **Scheduling window:** how far out to look for slots (Today only, Today and tomorrow, Next 3 days, Next 7 days).
- **Time slot selection:** Earliest available / Latest available.
- **Event duration:** default block size on the calendar (15/30/45/60 min).
- *(Coming soon)* **Booking link:** send clients a booking link and coordinate meetings with team members.

DECISIONING MODEL

Same tech-selection logic as Assign, plus a scheduling pass that finds an open slot on the picked tech's calendar within the configured window, respecting working hours, offset, and interval.

CONSTRAINTS

- Same team-uniqueness constraint as Assign (one Assign-or-Schedule agent per team).
- "Max active tickets per tech" is shown in Assign mode only — Schedule uses the calendar as the natural cap.

Cross-mode constraints (enforced in the prototype)

1. **One Route agent globally** per workspace.
2. **One Assign-tier agent per team** — a team can have either an Assign agent or an Assign + Schedule agent, but not both, and not two of either.
3. **A routing rule cannot dispatch to the board it's routing from.**
4. **One team = one board** assumption simplifies conflict detection; the constraint will need to relax when shared boards / company / TSI filters are added.

Domain primitives referenced

- **Board** — a queue/list of tickets in the PSA (e.g., ConnectWise PSA boards).
- **Team** — a group of technicians.
- **Tech / Technician** — an individual person who works tickets.
- **Status** — ticket lifecycle state (e.g., New, In Progress, Waiting on Client). Configured per workspace.
- **Agreement / Configuration / Item / Subtype** — PSA-side ticket metadata that drives routing decisions.

UI behavior notes

- **Mode picker** is a modal that lists the three modes as tiers; if a Route agent already exists, that option is hidden.
- **Agent list** sorts the Route agent (if any) on top, then Assign-tier agents.
- **Each rule's prompt** field is the only natural-language surface where admins teach the agent business logic. The "Agent context" hover panel surfaces the available ticket fields so admins know the agent already has that info implicitly.
- **Status multi-select** has a Select all / Clear all toggle that stays pinned at the bottom of the menu and a viewport-aware max height.

What's intentionally not in the prototype yet

- Shared boards across multiple teams (e.g. company / TSI-based filters).
- Booking link coordination (Assign + Schedule).
- SLA risk, Skills match, and Client familiarity weighting (Assign / Schedule).
- The structured `When...` filter editor for routing rules (code is preserved but commented out — may return alongside or instead of the natural-language prompt).
- Live ticket simulation / preview ("if you had this rule yesterday, here's what would have changed").